Solution

Exercises: Difference Equation Descriptions for Systems

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1. A difference equation describes the output of a linear time-invariant system as (a weighted sum, a weighted product, a convolution) of past values of (the output, cosines, impulses) and past/present values of (exponentials, the input).

2. The (coefficients, initial conditions, input) for the difference equation summarize all the past information about the system that is needed to determine its future.

3. Difference equations are widely used in (computing, integrating, differentiating) the output of LTI systems in signal processing, such as a discrete-time filter.

4. The transient response of a system described by a difference equation is determined by the (roots of the characteristic equation, roots of a Fredholm equation, derivative of the input signal).

5. The steady-state response of an LTI system described by a difference equation (takes the same form as the input, depends on initial conditions, gives the impulse response).

6. Which of the following correspond to linear, constant coefficient difference equations? Select all that apply.

a) $y[n] + 0.5y[n-1] + 0.25y[n-2] = x[n] + 2x[n-1]$

b) $y[n] + 0.5y[n-1]y[n-2] = x[n] + 2x[n-1]$ nonlinear due to $y[n-1]y[n-2]$

c) $y[n] = y[n-1] + 0.25y[n-2] - x[n]$

d) $y[n] + n^2y[n-1] + 0.25y[n-2] = 2x[n-1]$ not constant coeff

e) $y[n] + \cos(\pi n)y[n-1] + 0.25y[n-2] = x[n]x[n-1]$

f) $y[n] + 0.5y[n-1] + 0.1y[n-3] = x[n] - x[n-10]$

7. A system is described by the difference equation

$$y[n] - \frac{1}{2}y[n-1] = 2x[n-1]$$

a) Suppose that the system has the initial condition $y[-1] = -2$ and that no input is applied to the system. Find the first four values of the output, $y[0], y[1], y[2]$, and $y[3]$.

$x[n] = 0 \Rightarrow y[n] = \frac{1}{2} y[n-1]$

$y[0] = \frac{1}{2}(-2) = -1$

$y[1] = \frac{1}{2}(-1) = -\frac{1}{2}$

$y[2] = \frac{1}{2}(-\frac{1}{2}) = -\frac{1}{4}$

$y[3] = \frac{1}{2}(-\frac{1}{4}) = -\frac{1}{8}$

b) Recall that the impulse response of an LTI system is the output of the system when an impulse, $x[n] = \delta[n]$ is applied to a system at rest. Find the first four values of the impulse response $h[0], h[1], h[2]$, and $h[3]$.

at rest: $y[-1] = 0$, $x[n] = \delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$

$h[0] = \frac{1}{2}\overset{0}{y[-1]} + 2\overset{0}{x[-1]} = 0$

$h[1] = \frac{1}{2}\overset{0}{h[0]} + 2x[0] = 2$

$h[2] = \frac{1}{2}h[1] + 2\overset{0}{x[1]} = 1$

$h[3] = \frac{1}{2}h[2] + 2\overset{0}{x[2]} = \frac{1}{2}$

8. A system is described by the difference equation

$$y[n] + 0.25y[n-2] = x[n] + 3x[n-1]$$

a) Find the characteristic equation for this system.

$1 + 0.25z^{-2} = 0$

or $z^2 + 0.25 = 0$

b) Find the roots of the characteristic equation.

$\Rightarrow z^2 = -0.25$

$z = \pm j0.5$

9. A system is described by the difference equation

$$y[n] + 0.25y[n-2] = x[n] + 3x[n-1]$$

a) Recall that the system function is a ratio of polynomials, $H(z) = b(z)/a(z)$. Find the numerator polynomial $b(z)$.

$b(z) = 1 + 3z^{-1}$

b) Find the coefficients of the denominator polynomial $a(z)$.

$a(z) = 1 + 0.25z^{-2}$

c) Suppose the input to this system is $x[n] = 2u[n]$. Find the steady-state response of the system. Hint: express the input for large n as $x[n] = \alpha z^n$.

$x[n] = \alpha z^n \longrightarrow y[n] = \alpha H(z) z^n$

for steady state $x[n] = 2 \Rightarrow \alpha = 2, z = 1$

$y_s[n] = 2H(1) = 2\frac{1+3}{1+.25} = \frac{8}{1.25}$

d) Suppose the input to this system is $x[n] = 3(-1)^n u[n]$. Find the steady-state response of the system.

for steady state $x[n] = 3(-1)^n \Rightarrow \alpha = 3, z = -1$

$y_s[n] = 3H(-1)(-1)^n = 3\frac{(1-3)}{(1-0.25)}(-1)^n$

$= \frac{-6}{0.75}(-1)^n$

10. A system is described by the difference equation

$$y[n] - 0.25y[n-2] = x[n] + 3x[n-1]$$

Assume the system has initial conditions $y[-2] = 0, y[-1] = 4$ and that the input to the system is $x[n] = 0$.

a) Find the roots of the characteristic equation.

$1 - 0.25z^{-2} = 0$ or $z^2 - 0.25 = 0$

$z = \pm 0.5$

b) Translate the initial conditions forward in time to find $y[0], y[1]$.

$y[n] = 0.25\,y[n-2] + x[n] + 3x[n-1]$ but $x[n] = 0$

$y[0] = 0.25\,y[-2] = 0$

$y[1] = 0.25\,y[-1] = 1$

c) Find the output of the system for $n > 0$. Note that since the input $x[n] = 0$, the output has zero steady-state component.

$y[n] = C_1 (.5)^n + C_2 (-.5)^n \;, \quad n \geq 0$

$y[0] = 0 = C_1 + C_2$

$y[1] = 1 = \frac{1}{2} C_1 - \frac{1}{2} C_2$) $\times 2$ and add eqns

$2 = 2C_1 \Rightarrow C_1 = 1$

$\Rightarrow C_2 = -1$

$y[n] = \left(\frac{1}{2}\right)^n - \left(-\frac{1}{2}\right)^n \quad n \geq 0$

11. A system is described by the difference equation

$$y[n] - (1/2)y[n-1] = 2x[n]$$

Assume the input is $x[n] = (1/4)^n u[n]$ and the initial condition is $y[-1] = 1$.

a) Find the steady-state component of the system output.

under steady state $x[n] = (1/4)^n \Rightarrow y_s[n] = H(1/4)\,(1/4)^n$

$H(z) = \dfrac{2}{1 - 1/2\, z^{-1}}$ $\quad H(1/4) = \dfrac{2}{1 - 1/2(4)} = -2$

$y_s[n] = -2\,(1/4)^n \quad n \geq 0$

b) Find the transient component of the output.

char eqn $1 - 1/2\, z^{-1} = 0$

roots $z = 1/2$

$y[n] = C_1 (1/2)^n - 2(1/4)^n \quad n \geq 0$

need $y[0]$: $y[n] = 1/2\, y[n-1] + 2x[n]$

$y[0] = \frac{1}{2}(1) + 2 \cdot 1 = 2.5$

So $2.5 = C_1 - 2 \Rightarrow C_1 = 4.5$

$y[n] = \underbrace{4.5\,(1/2)^n}_{y_t[n] = 4.5(1/2)^n \quad n \geq 0} - 2(1/4)^n \quad n \geq 0$

12. A system is described by the difference equation

$$y[n] - 0.25y[n-1] - 0.25y[n-2] + (1/16)y[n-3] = x[n] + \beta x[n-1] + \gamma x[n-3]$$

where β and γ are unknown constants. The input to the system is $x[n] = \cos((\pi/4)n)u[n]$. Of the following candidate steady-responses, which could NOT be correct? Select all that apply.

a) $y_s[n] = 2\sin((\pi/4)n)u[n]$

b) $y_s[n] = 2\cos((\pi/8)n)u[n]$

c) $y_s[n] = [2\cos((\pi/4)n) - \sin((\pi/4)n)]u[n]$

d) $y_s[n] = 2(\pi/4)^n u[n]$

e) $y_s[n] = 2\sin((\pi/2)n)u[n]$

f) $y_s[n] = (\pi/4)u[n]$

$$H(z) = \frac{1 + \beta z^{-1} + \gamma z^{-3}}{1 - 0.25z^{-1} - 0.25z^{-2} + \frac{1}{16}z^{-3}}$$

steady state: $x[n] = \cos\left(\frac{\pi}{4}n\right)$

$$= \frac{1}{2}e^{j\pi/4 n} + \frac{1}{2}e^{-j\frac{\pi}{4}n}$$

$$= \frac{1}{2}(z_1)^n + \frac{1}{2}(z_2)^n$$

$$z_1 = e^{j\pi/4}, \quad z_2 = e^{-j\pi/4}$$

So $y_s[n] = \frac{1}{2}H(z_1)e^{j\pi/4 n} + \frac{1}{2}H(z_2)e^{-j\pi/4 n}$

$H(z)$ is unknown, but $y_s[n]$ only involves $e^{j\pi/4 n}$ and $e^{-j\pi/4 n}$ → any valid output can only have these terms. a) and c) are only possible valid outputs

13. Which of the following systems described by difference equations is likely to have the fastest transient response?

a) $y[n] - (9/16)y[n-2] = x[n]$

b) $y[n] + 0.1y[n-1] = 4x[n]$

c) $y[n] + 0.81y[n-2] = 0.1x[n]$

d) $y[n] - 0.5y[n-1] = 0.5x[n]$

transient response depends on roots of characteristic equation

root $d_i \longrightarrow d_i^n$ fast response $\Rightarrow$ $|d_i|$ small

a) roots $d_{1,2} = \pm 3/4$

b) root $d_1 = -0.1$

c) roots $d_{1,2} = \pm j0.9$

d) root $d_1 = 0.5$

transient response decays fastest for case b!